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BSCCS/2024/55349

BSC Computer Science

Unit Code: BIT3208 Advanced Web Design and Development

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May/August 2025/2026

Project Details

Project Title: Runi Empire Grooming Barbershop Management System

Selected Technology: JavaScript, PHP, MySQL, XAMPP

GitHub Link:

https://github.com/deniskibichiy/BIT3208_Advanced_Web_Design_and_Development_BSCCS_2024_55349

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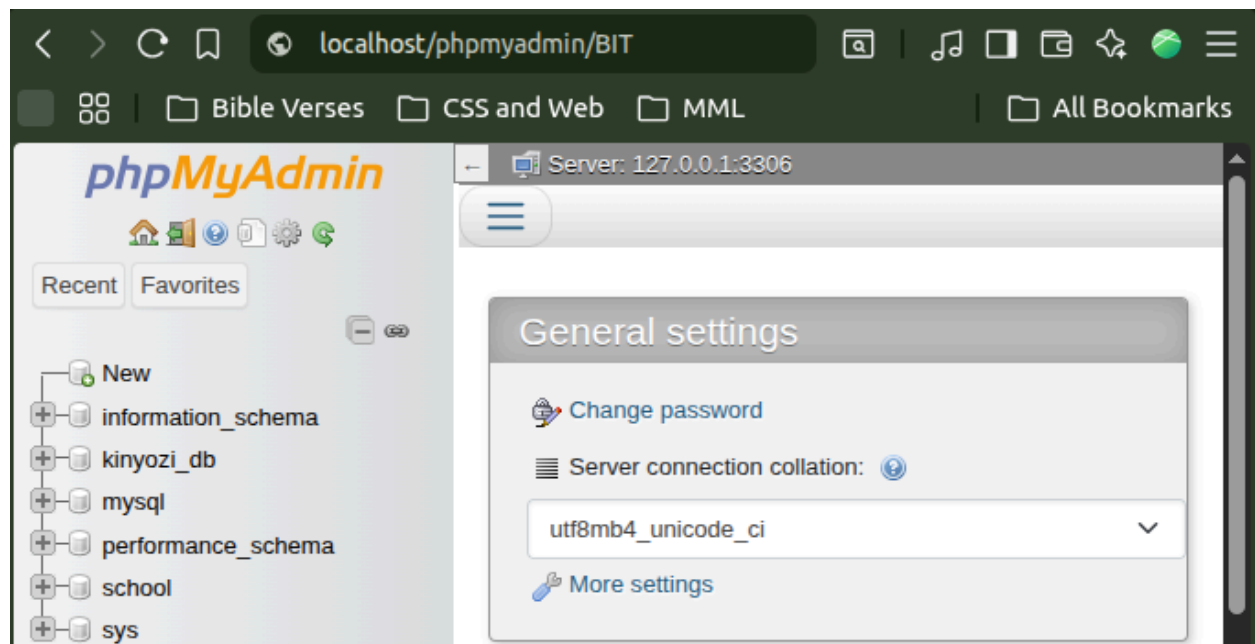
1. Week 1: Environment setup and database connection

1.1. Figure 1: XAMPP Installation and Setup with MySQL in Linux

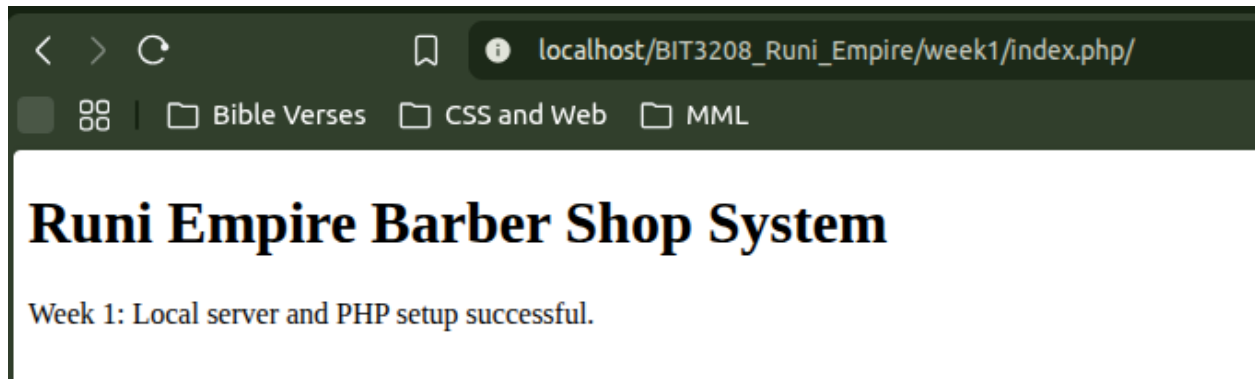
```
denis@denis-HP-ProBook-450-G7:~$ sudo /opt/lampp/xampp status
Version: XAMPP for Linux 8.2.12-0
Apache is running.
MySQL is not running.
ProFTPD is not running.
denis@denis-HP-ProBook-450-G7:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/lib/systemd/system/mysql.service; enabled; vendor p
   Active: active (running) since Thu 2026-05-28 13:18:55 EAT; 39min ago
   Main PID: 10633 (mysqld)
   Status: "Server is operational"
   Tasks: 39 (limit: 18907)
   Memory: 434.2M
   CPU: 14.863s
   CGroup: /system.slice/mysql.service
           └─10633 /usr/sbin/mysqld

May 28 13:18:53 denis-HP-ProBook-450-G7 systemd[1]: Starting MySQL Commun
May 28 13:18:55 denis-HP-ProBook-450-G7 systemd[1]: Started MySQL Communi
denis@denis-HP-ProBook-450-G7:~$ ss -tulnp | grep 3306
tcp    LISTEN 0      70      127.0.0.1:3306  0.0.0.0:*
tcp    LISTEN 0      151     127.0.0.1:3306  0.0.0.0:*
denis@denis-HP-ProBook-450-G7:~$
```

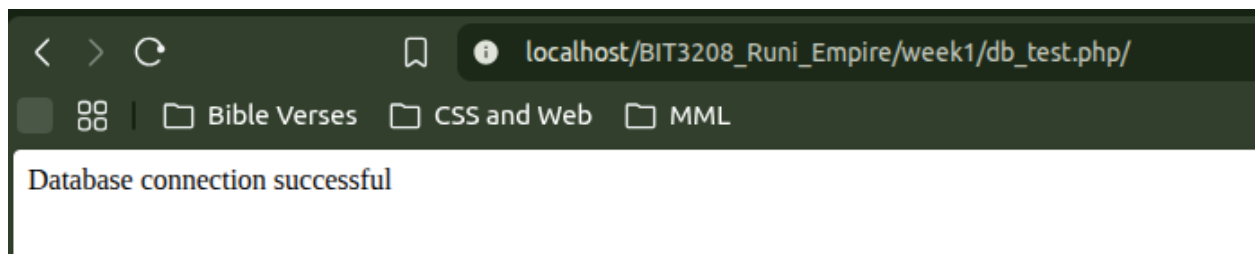
1.2. Figure 2: Apache + MySQL Running and Operational



1.3. Figure 3: Index.php successfully rendered



1.4. Figure 4: Database connection successful



1.5. Figure 5: Week1 database

```
mysql> USE week1db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_week1db |
+-----+
| users              |
+-----+
1 row in set (0.01 sec)

mysql> DESCRIBE users
-> ;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int  | NO   | PRI | NULL    | auto_increment |
| name  | varchar(100) | YES |     | NULL    |
| phone_number | varchar(20) | YES |     | NULL    |
| created_at | timestamp | YES |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

Week 1 reflection:

During week one, the focus was on setting up the environment to facilitate practical activities which are part of the Unit requirements. A key challenge I experienced is that the XAMPP MySQL couldn't load successfully due to port clashes with the system MySQL. To solve this, I had to configure Apache to use system MySQL instead of the one provided by PHPMyAdmin and the process was successful.

2. Week 2: User Interface Planning and System Design

2.1. Figure 2.1: Homepage User Interface Mockup



2.2. Figure 2.2: Login Page User Interface Mockup

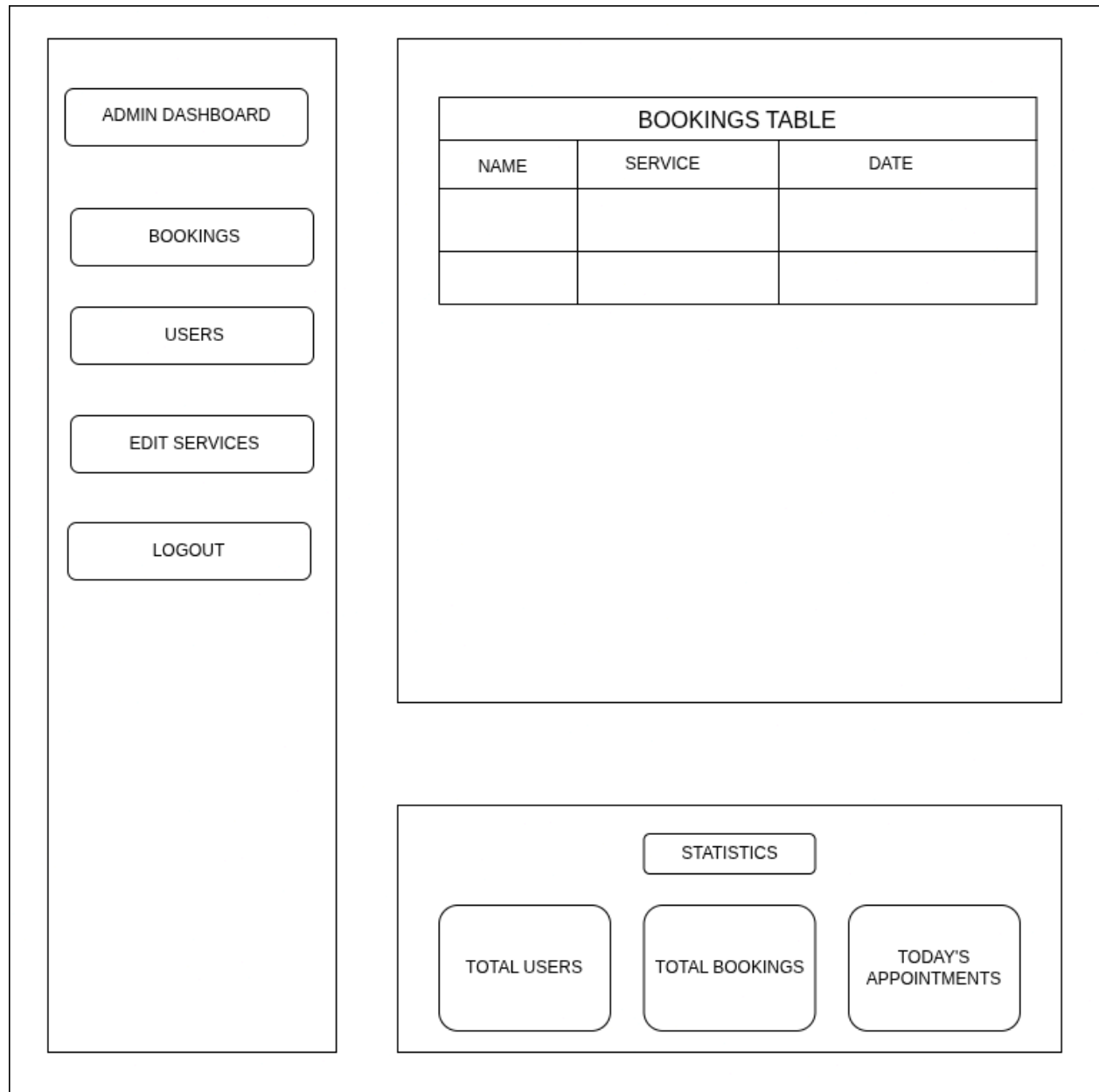
WELCOME TO RUNI EMPIREE

Username:

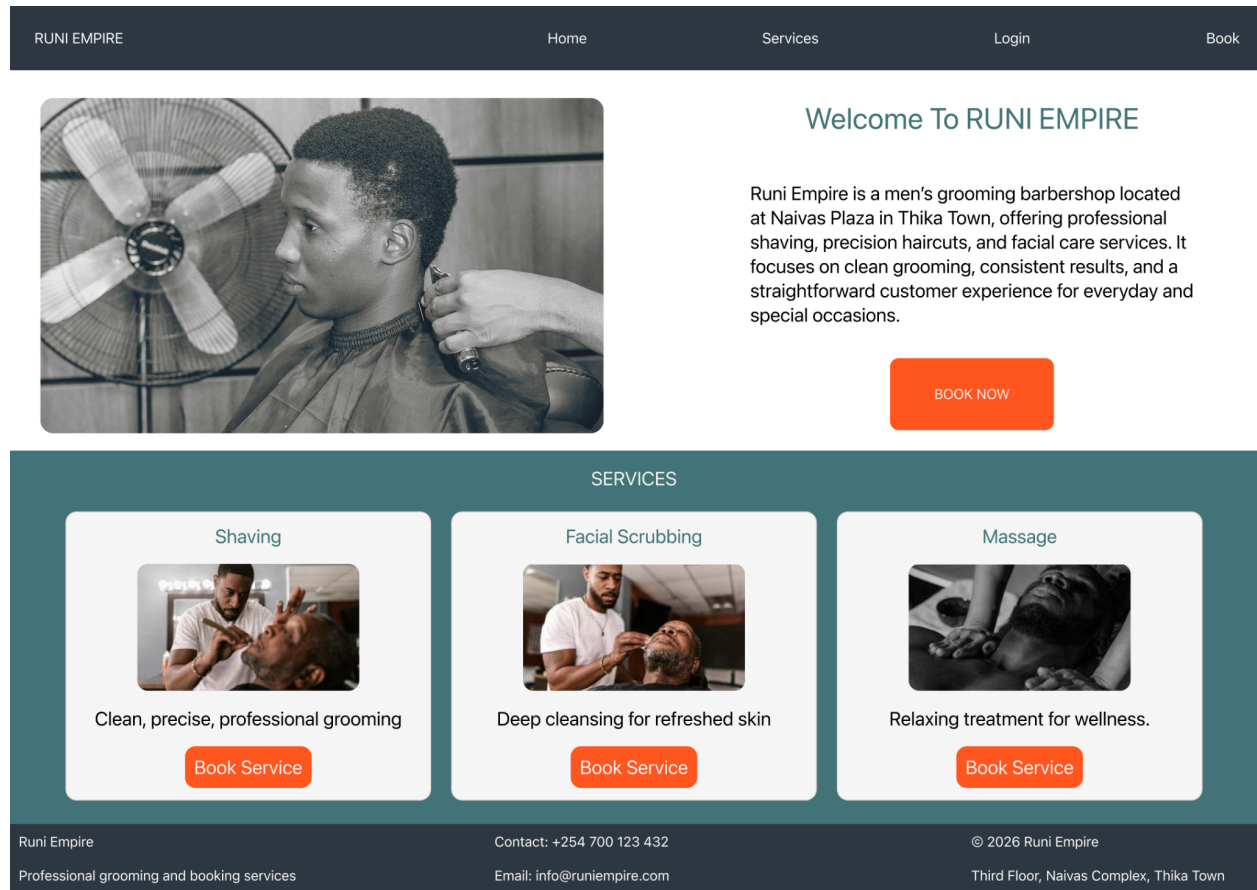
Password:

LOGIN

This is a user interface mockup for a login page. It features a rectangular container with a thin black border. Inside, the text 'WELCOME TO RUNI EMPIREE' is centered at the top. Below this, the label 'Username:' is followed by a horizontal text input field. The label 'Password:' is followed by a horizontal password input field. At the bottom, there is a wide rectangular button with the text 'LOGIN' centered on it.

2.3. Figure 2.3: Admin Dashboard User Interface Mockup

2.4. Figure 2.1: Homepage Figma prototype



2.5. Login page Figma prototype

RUNI EMPIRE

Home

Services

Login

Book

Login to Access Our Services

Email:

Password:

Login

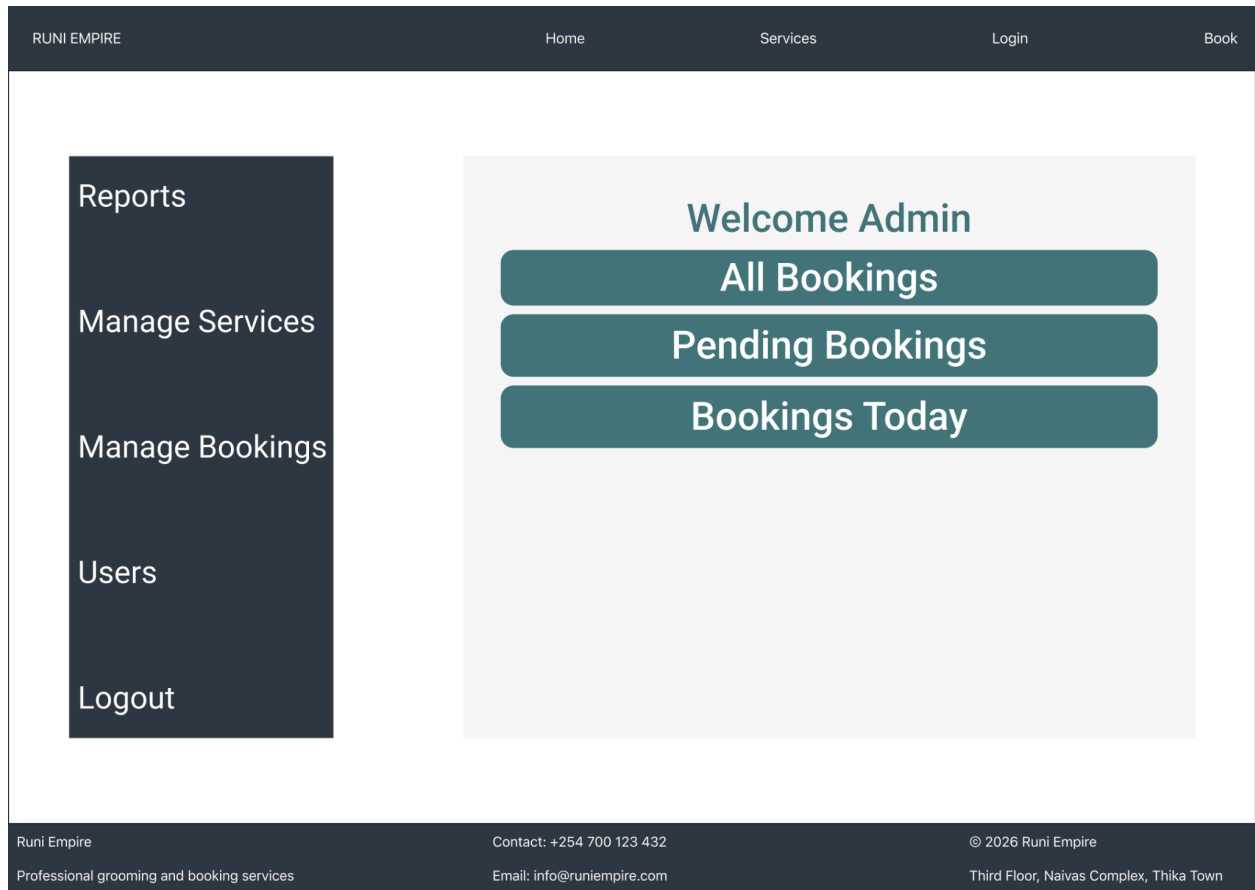
Don't have an account? [Sign Up](#)

Runi Empire
Professional grooming and booking services

Contact: +254 700 123 432
Email: info@runiempire.com

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Third Floor, Naivas Complex, Thika Town

2.6. Dashboard Figma prototype

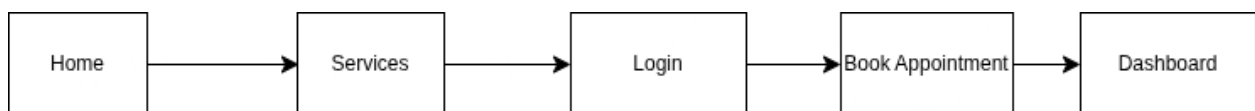


2.7. Figure 2.8: Color Scheme



- i. Background: #F5F5F5 : Light Gray
- ii. Secondary/Highlights: #76ABAE: Teal
- iii. Main text: #303841: Charcoal
- iv. Accent/Actions: #FF5722: Orange

2.8. Figure 2.9: Navigation Structure



Reflection week 2:

Week 2 provided me with another key challenge in that after visualizing how my website should be laid out, the next obstacle was designing the prototype using Figma. To complete the process

successfully, I had to consider carefully how to come up with reusable components to avoid creating frames repetitively.

3. JavaScript and PHP Basics

3.1. Figure 1: Invalid Login Validation

Create Account to Access our Services

Name

Email

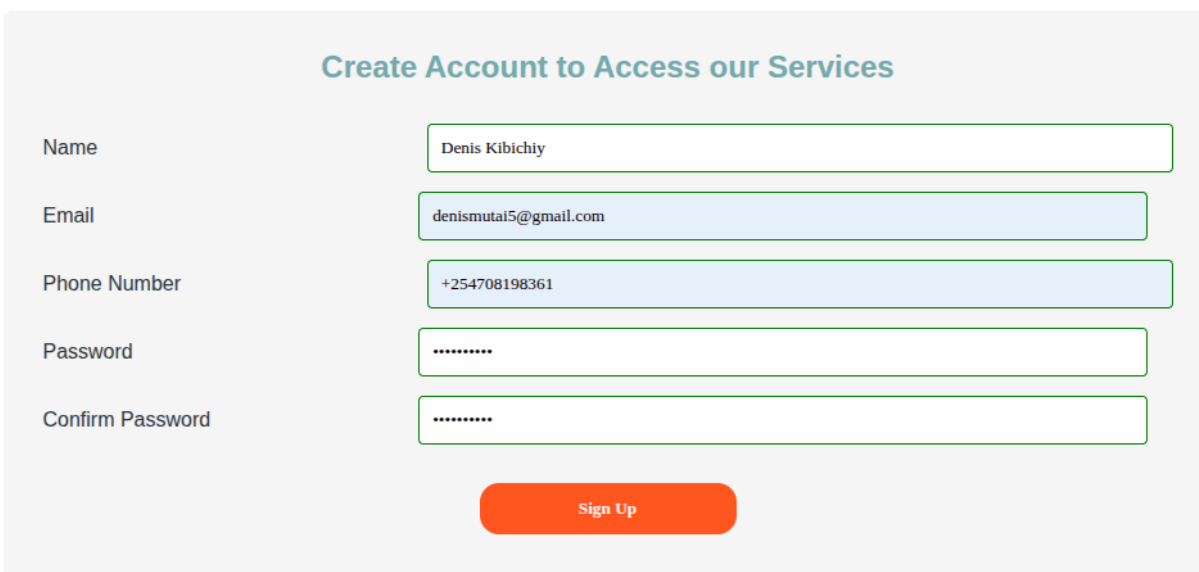
Phone Number

Password

Confirm Password Passwords must match

Already have an account? [Login](#)

3.2. Fig 2: Valid login input accepted



The screenshot shows a 'Create Account to Access our Services' form. It contains five input fields: Name (Denis Kibichiy), Email (denismutai5@gmail.com), Phone Number (+254708198361), Password (masked with dots), and Confirm Password (masked with dots). A blue 'Sign Up' button is at the bottom.

Create Account to Access our Services

Name Denis Kibichiy

Email denismutai5@gmail.com

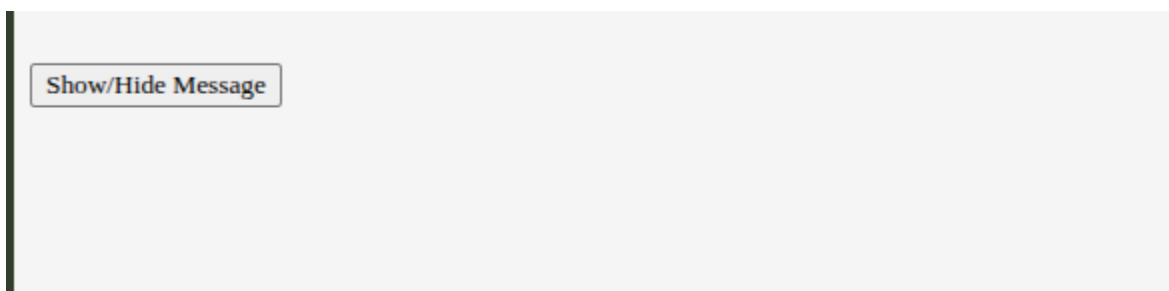
Phone Number +254708198361

Password

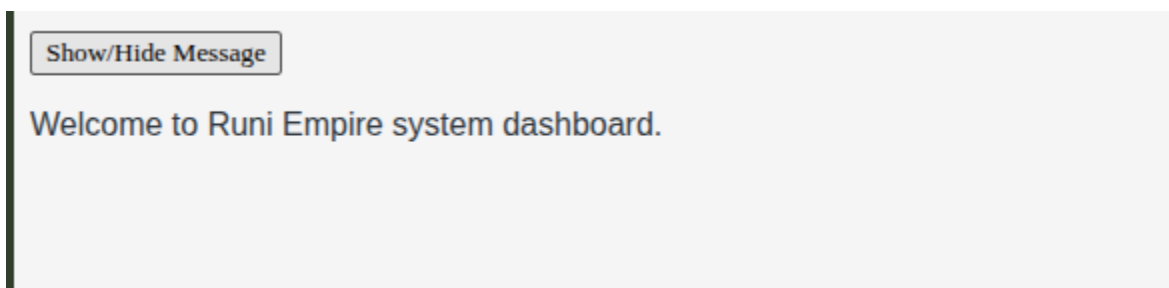
Confirm Password

Sign Up

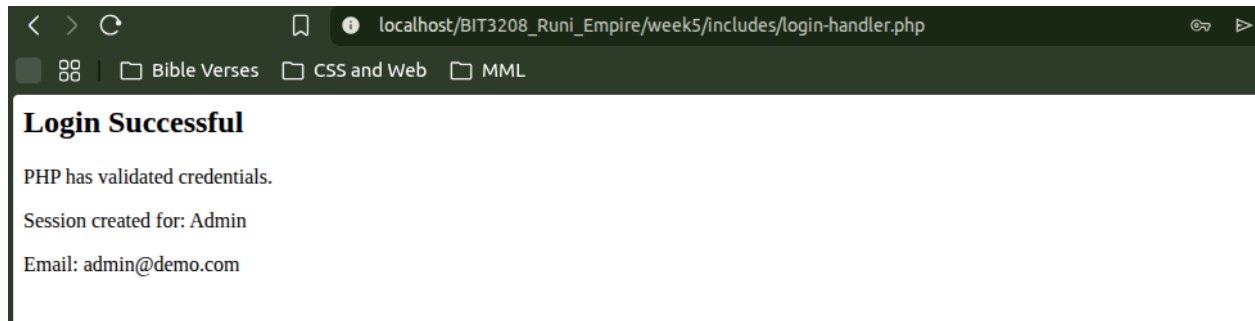
3.3. Fig 3: DOM manipulation interaction



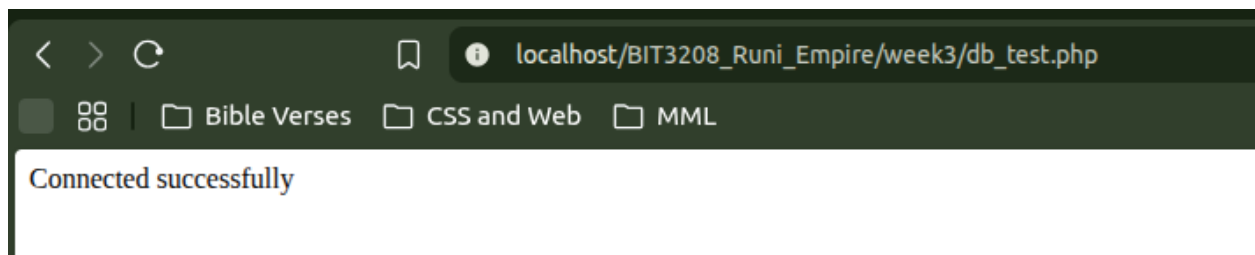
After button click



3.4. Figure 4: PHP Interaction Evidence after valid login details captured



3.5. Figure 5: Week 3 database connection success evidence

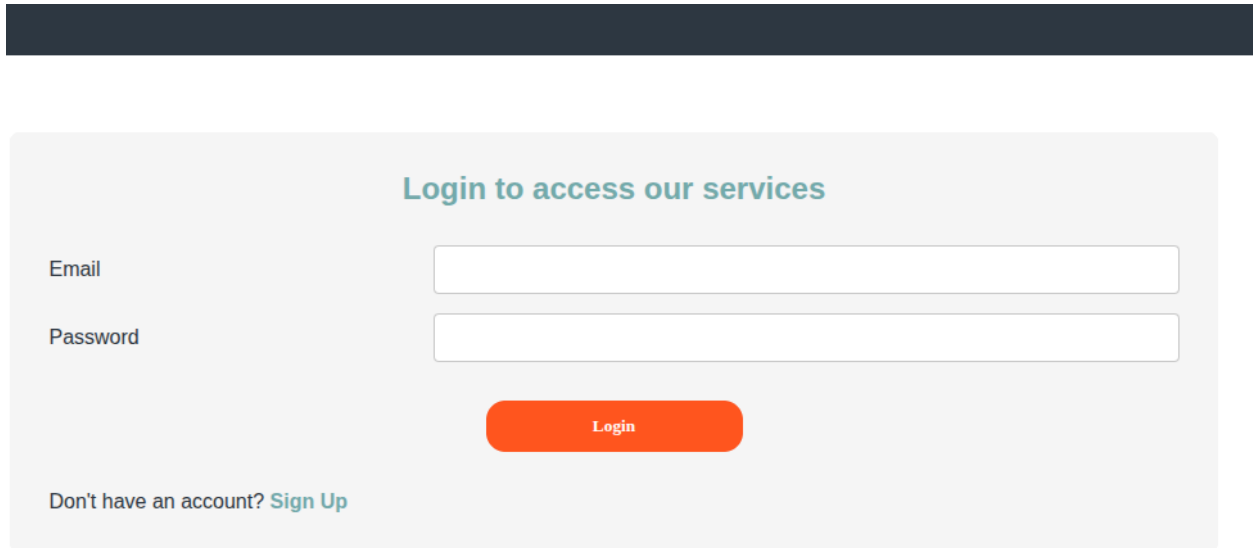


Week 3 Reflection:

In week 3, much of the focus shifted to JavaScript and PHP processing. The most challenging aspect was using JavaScript to perform form validation, and to succeed, I had to learn a lot of concepts, mostly dealing with DOM Manipulation such as query selectors, creating nodes, and appending them to already existing parent containers.

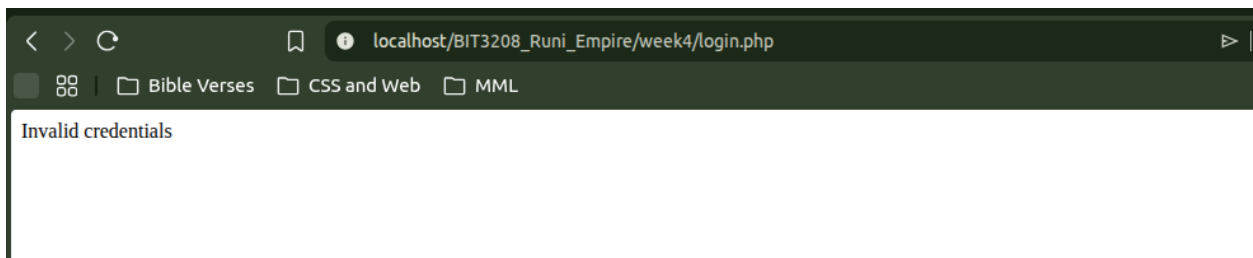
4. WEEK 4: Server-Side Components and Backend Development Foundations

4.1. Figure 1: Login page initial state

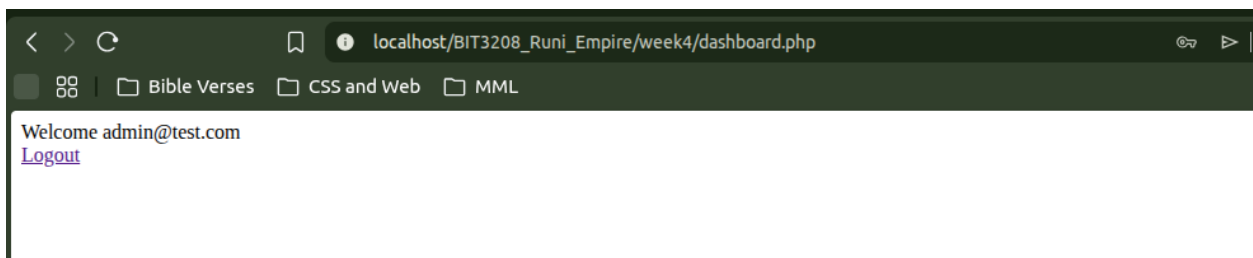


The image shows a login form on a light gray background. At the top, the text "Login to access our services" is displayed in a teal color. Below this, there are two input fields: "Email" and "Password". The "Email" field is a white rectangle with a thin gray border. The "Password" field is a white rectangle with a thin gray border. Below the "Password" field is an orange rounded rectangle button with the text "Login" in white. At the bottom left, there is a link that says "Don't have an account? Sign Up" in a teal color.

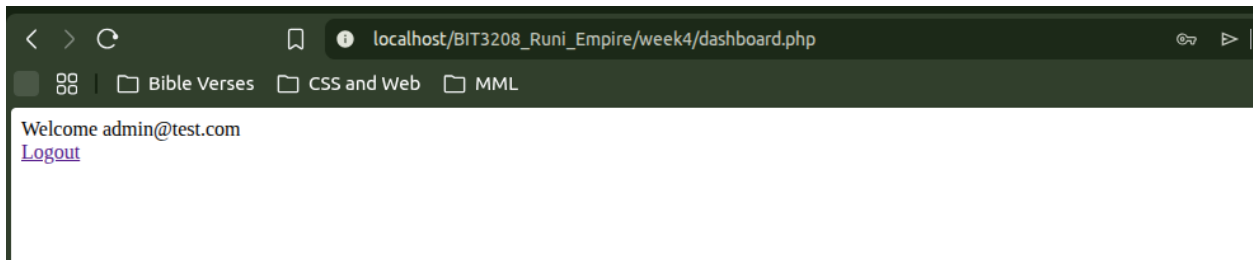
4.2. Figure 2: Failed Login Attempt



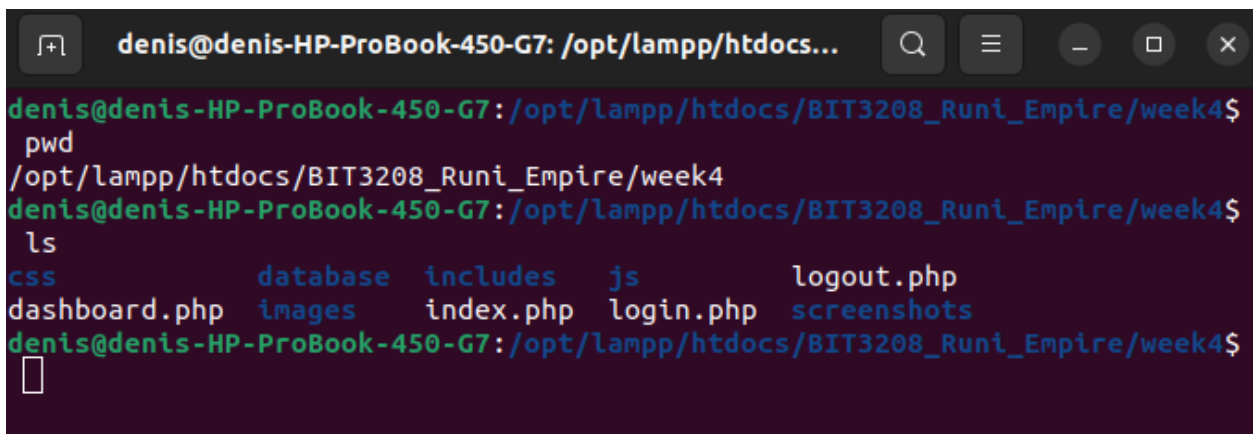
4.3. Figure 3: Successful login attempt - Redirected to dashboard



4.4. Figure 4: Session-based admin dashboard



4.5. Figure 5: Folder Structure



Week 4 Reflection:

In Week 4, the focus was on form processing, authentication basics, and sessions in PHP. The most difficult part was understanding how data moves from HTML forms into PHP and how validation is applied before processing. I learned how to capture form input using POST, validate it, and manage user sessions to maintain login state. I also built a basic login system and session-protected dashboard. This helped me understand how authentication systems control access to different parts of a web application.

5. Week 5: Database Components and CRUD Operations

5.1. Figure 5.1: Databases Screenshot

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| kinyozi_db |
| mysql |
| performance_schema |
| runi_empire_db |
| school |
| sys |
| week1db |
| week_one |
+-----+
9 rows in set (0.01 sec)
```

5.2. Figure 5.2: Created Tables within Database and defined their schema

```
mysql> USE runi_empire_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_runi_empire_db |
+-----+
| bookings |
| users |
+-----+
2 rows in set (0.00 sec)

mysql> DESCRIBE users;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id | int | NO | PRI | NULL | auto_increment |
| first_name | varchar(100) | YES | | NULL | |
| last_name | varchar(100) | YES | | NULL | |
| email | varchar(100) | YES | UNI | NULL | |
| phone_number | varchar(20) | YES | | NULL | |
| password | varchar(255) | YES | | NULL | |
| created_at | timestamp | YES | | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)
```

5.3. Figure 5.3: CRUD Operations on database

```
mysql> INSERT INTO users (first_name, last_name, email, phone_number, password)
-> VALUES ('Brian', 'Kiptoo', 'brian.kiptoo@gmail.com', '0712345678', '1234');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM users;
+-----+-----+-----+-----+-----+-----+-----+
| id | first_name | last_name | email | phone_number | password | created_at |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Brian | Kiptoo | brian.kiptoo@gmail.com | 0712345678 | 1234 | 2026-06-05 11:47:07 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> UPDATE users
-> SET first_name = 'Kevin',
-> last_name = 'Mutua'
-> WHERE id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

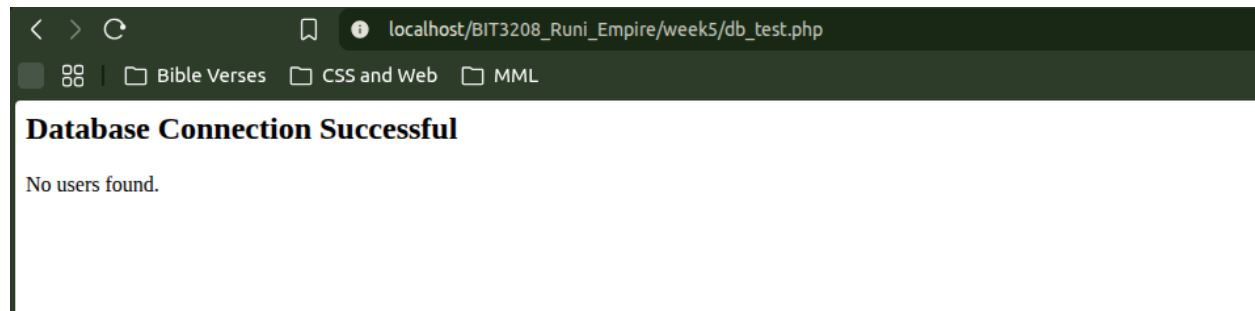
mysql> SELECT * FROM users;
+-----+-----+-----+-----+-----+-----+-----+
| id | first_name | last_name | email | phone_number | password | created_at |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Kevin | Mutua | brian.kiptoo@gmail.com | 0712345678 | 1234 | 2026-06-05 11:47:07 |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> DELETE FROM users
-> WHERE id = 1;
Query OK, 1 row affected (0.01 sec)

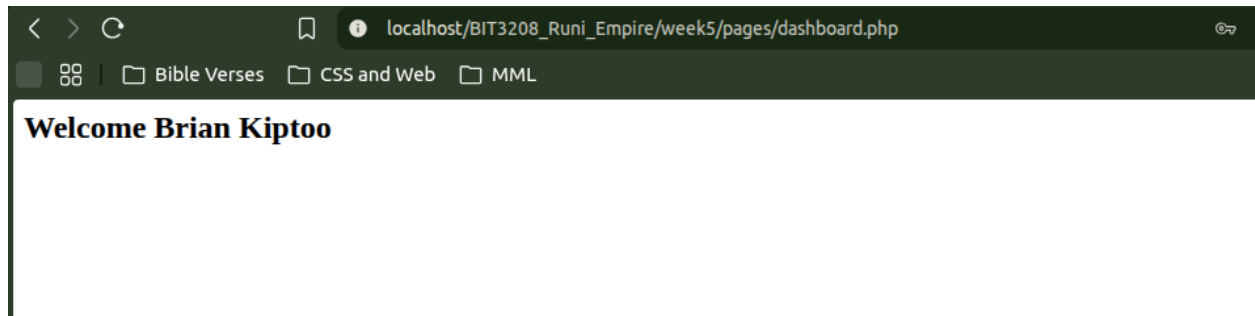
mysql> SELECT * FROM users;
Empty set (0.00 sec)

mysql> 
```

5.4. Figure 5.4: Database Connection



5.5. Figure 5.5: Successful login through the database



Week 5 Reflection:

In Week 5, I focused on databases and integrating them with PHP applications. The main challenge was understanding SQL operations and how PHP interacts with MySQL databases. I learned how to create databases and tables, perform CRUD operations, and connect PHP to a database using PDO. I also implemented database-driven login functionality and data retrieval. This improved my understanding of backend development and data management in web applications.